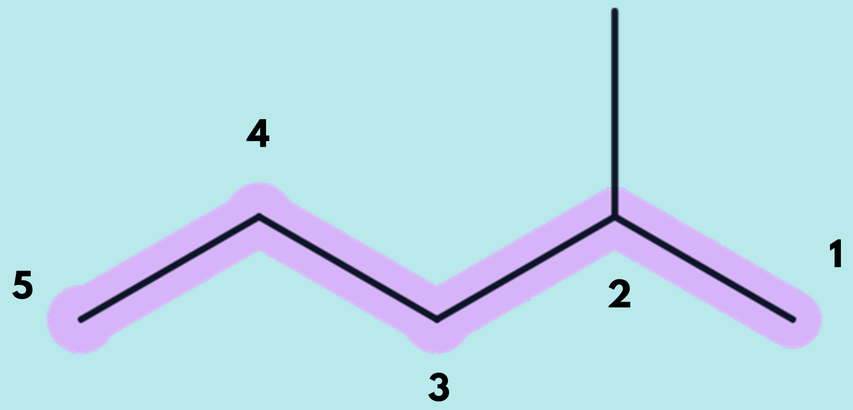
CARD A01

Level: Foundation | Molecular Formula: C_6H_{14}

NUMBERED PARENT



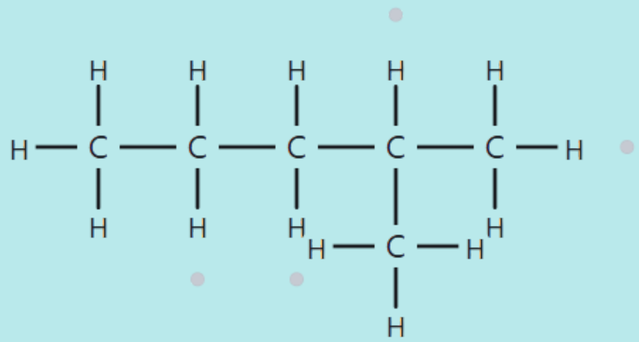
NAMING TRAIL

Parent: Longest chain = 5C → pentane
Numbering: left → right gives 4; right → left gives 2.
First point of difference: 2 < 4, so number from the right.
Substituents: Methyl on C2 → 2-methyl

MY NAME

2-methylpentane

THE ALKANE CHECK



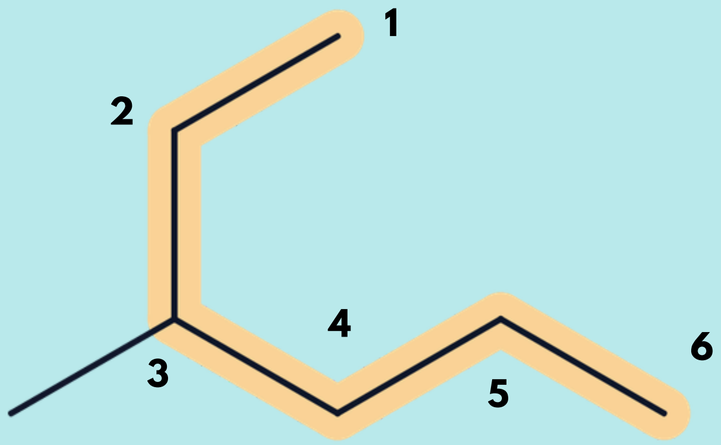
Returned Name: 2-methylpentane

☑ LONGEST CHAIN ☑ LOWEST LOCANTS ☑ ALPHABETICAL ORDER ☑ PUNCTUATION ☑ FORMULA COUNTED

CARD A02

Level: Foundation | Molecular Formula: C_7H_{16}

NUMBERED PARENT



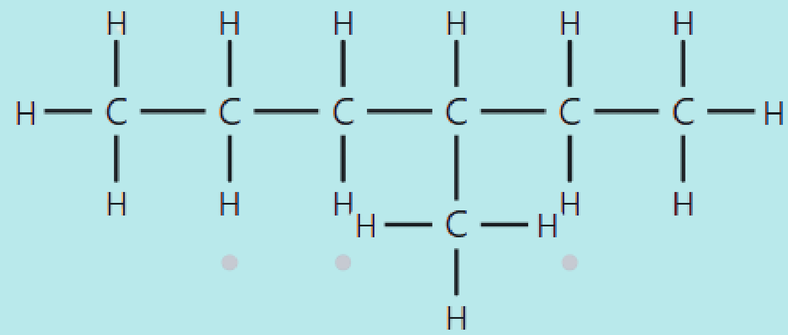
NAMING TRAIL

Parent: Longest chain = 6C → hexane
Numbering: left → right gives 4; right → left gives 3.
First point of difference: 3 < 4, so number from the right.
Substituents: Methyl on C3 → 3-methyl

MY NAME

3-methylhexane

THE ALKANE CHECK



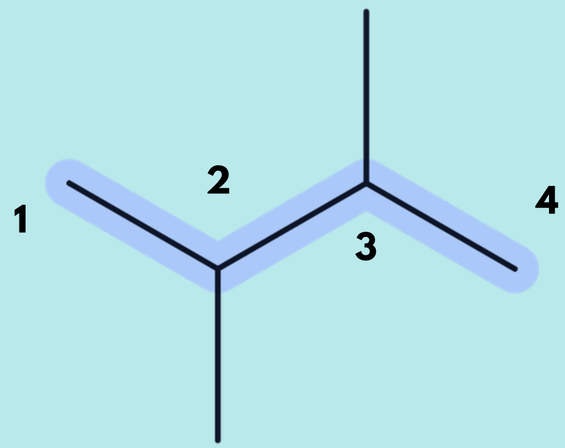
Returned Name: 3-methylhexane

☑ LONGEST CHAIN ☑ LOWEST LOCANTS ☑ ALPHABETICAL ORDER ☑ PUNCTUATION ☑ FORMULA COUNTED

CARD A03

Level: Foundation | Molecular Formula: C_6H_{14}

NUMBERED PARENT



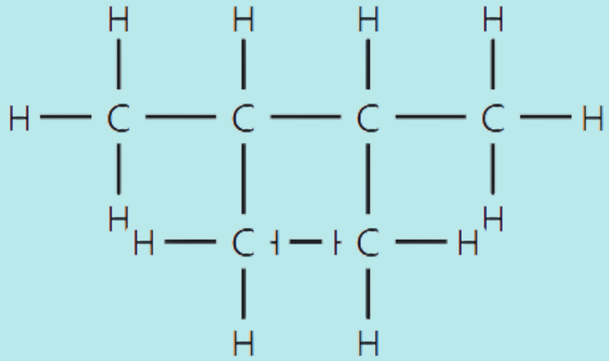
NAMING TRAIL

Parent: Longest chain = 6C → hexane
Numbering: left → right gives 2, 3; right → left gives 2,3.
Symmetrical, so either direction leads to the same name.
Substituents: Methyl groups on C2 and C3 → 2,3-dimethyl

MY NAME

2,3-dimethylbutane

THE ALKANE CHECK



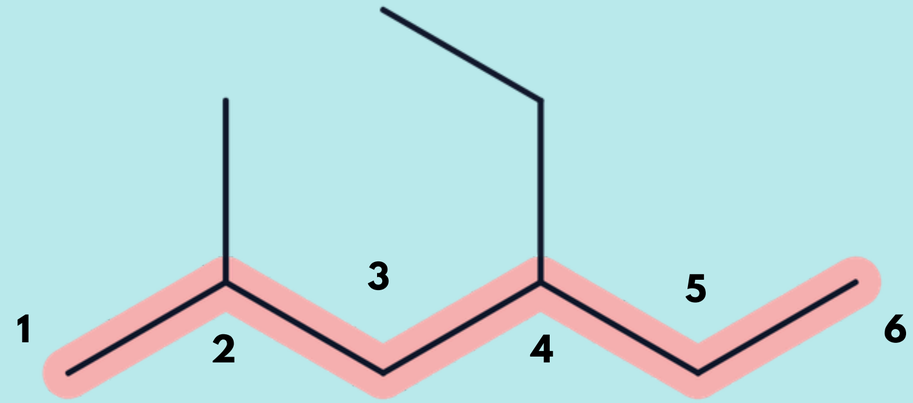
Returned Name: 2,3-dimethylbutane

☑ LONGEST CHAIN ☑ LOWEST LOCANTS ☑ ALPHABETICAL ORDER ☑ PUNCTUATION ☑ FORMULA COUNTED

CARD A04

Level: Developing | Molecular Formula: C_9H_{21}

NUMBERED PARENT



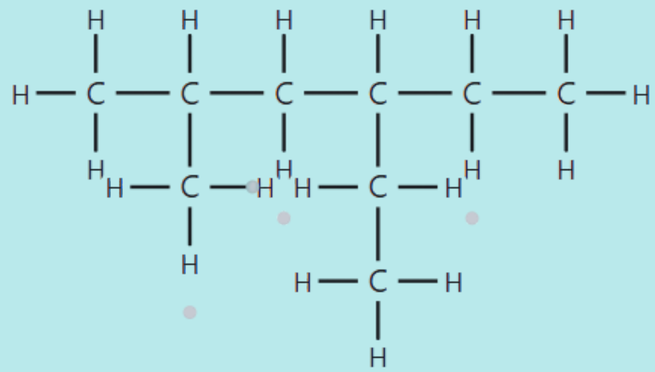
NAMING TRAIL

Parent: Longest chain = 6C → hexane
Numbering: left → right gives 2,4; right → left gives 3,5
First point of difference: 2 < 3, so number from the left.
Substituents: Methyl on C2, Ethyl on C4 → 4-ethyl, 2-methyl

MY NAME

4-ethyl-2-methylhexane

THE ALKANE CHECK



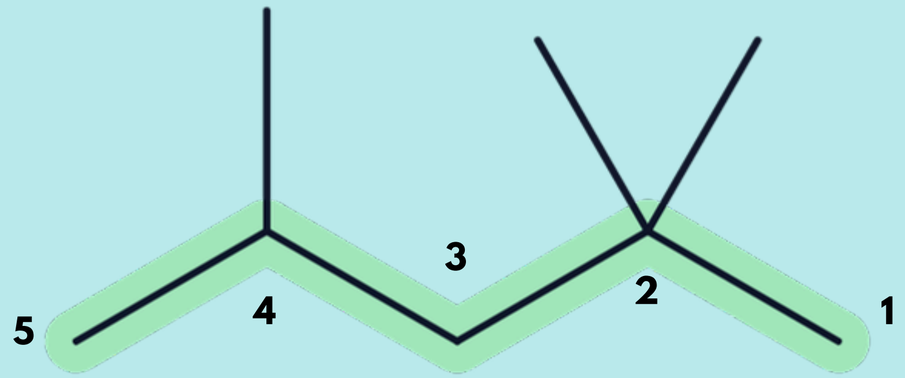
Returned Name: 4-ethyl-2-methylhexane

☑ LONGEST CHAIN ☑ LOWEST LOCANTS ☑ ALPHABETICAL ORDER ☑ PUNCTUATION ☑ FORMULA COUNTED

CARD A05

Level: Developing | Molecular Formula: C_8H_{18}

NUMBERED PARENT



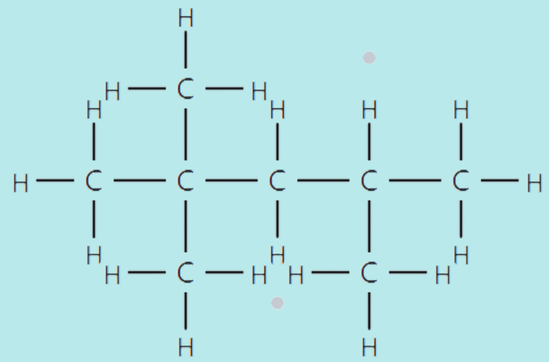
NAMING TRAIL

Parent: Longest chain = 6C → hexane
Numbering: left → right gives 2,4,4; right → left gives 2,2,4.
Second point of difference: 2 < 4, so number from the left.
Substituents: Methyl groups on C2 and C4 → 2,2,4-trimethyl

MY NAME

2,2,4-trimethylpentane

THE ALKANE CHECK



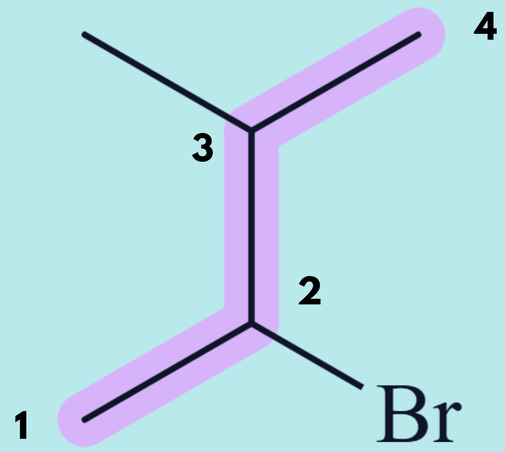
Returned Name: 2,2,4-trimethylpentane

☑ LONGEST CHAIN ☑ LOWEST LOCANTS ☑ ALPHABETICAL ORDER ☑ PUNCTUATION ☑ FORMULA COUNTED

CARD A06

Level: Developing | Molecular Formula: C_5H_{11}

NUMBERED PARENT



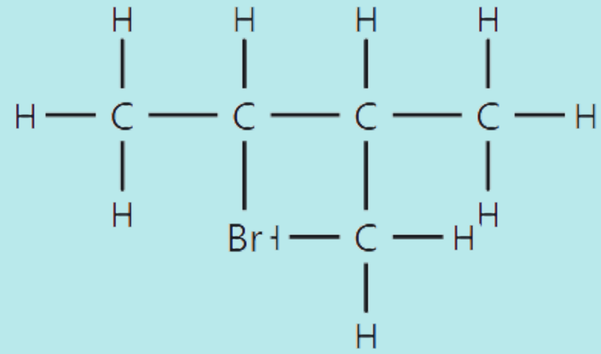
NAMING TRAIL

Parent: Longest chain = 4C → butane
Numbering: bottom left → top right gives 2,3; top right → bottom left gives 2,3. Tie-breaker: alphabetical priority means bromo gets the lower number (C2) over methyl (C3).
Substituents: Bromo on C2, Methyl on C3 → 2-bromo-3-methyl

MY NAME

2-bromo-3-methylbutane

THE ALKANE CHECK



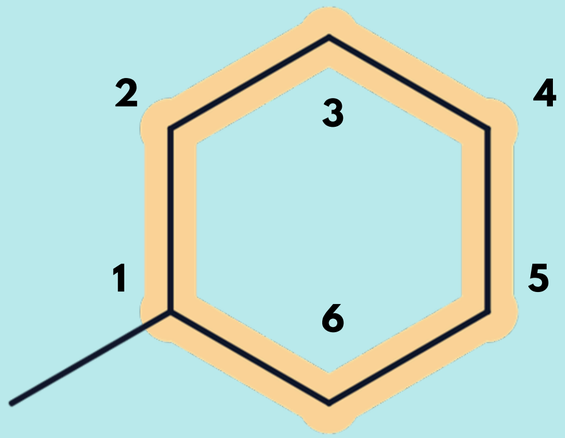
Returned Name: 2-bromo-3-methylbutane

☑ LONGEST CHAIN ☑ LOWEST LOCANTS ☑ ALPHABETICAL ORDER ☑ PUNCTUATION ☑ FORMULA COUNTED

CARD A07

Level: Developing | Molecular Formula: C_7H_{14}

NUMBERED PARENT



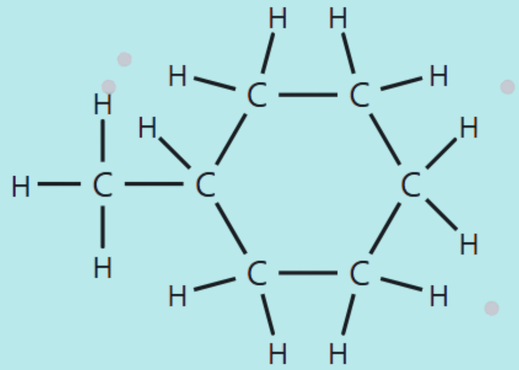
NAMING TRAIL

Parent: 6C ring → cyclohexane
Numbering: With only one substituent on a ring, the attached carbon is automatically C1
Substituents: Methyl on C1 → 1-methyl

MY NAME

1-methylcyclohexane

THE ALKANE CHECK



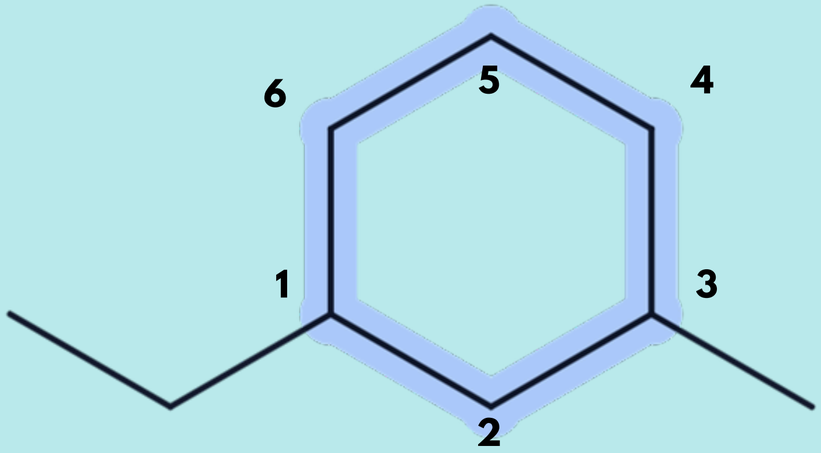
Returned Name: 1-methylcyclohexane

☑ LONGEST CHAIN ☑ LOWEST LOCANTS ☑ ALPHABETICAL ORDER ☑ PUNCTUATION ☑ FORMULA COUNTED

CARD A08

Level: Challenge | Molecular Formula: C_9H_{18}

NUMBERED PARENT



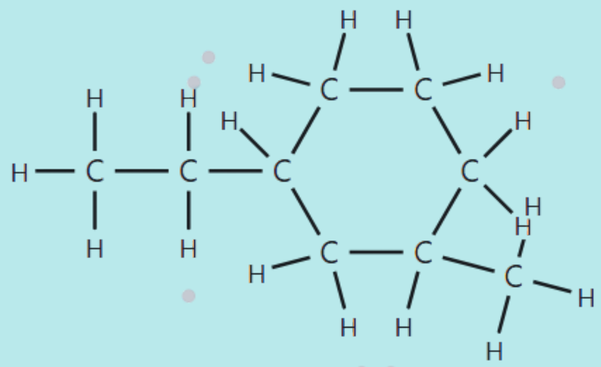
NAMING TRAIL

Parent: 6C ring → cyclohexane
Numbering: Always number around the ring in the direction that gives the substituents the lowest possible numbers. Start at the ethyl group (alphabetical priority over methyl) to make it C1. Number towards the methyl group to give it the lowest possible locant (C3)
Substituents: Ethyl on C1, Methyl on C3 → 1-ethyl-3-methyl

MY NAME

1-ethyl-3-methylcyclohexane

THE ALKANE CHECK



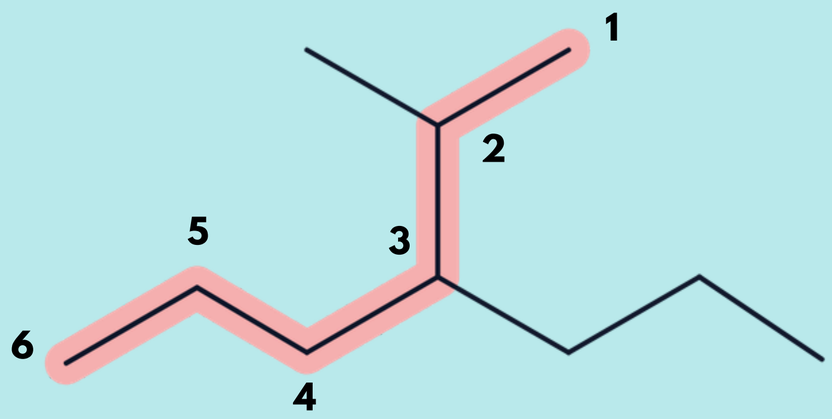
Returned Name: 1-ethyl-3-methylcyclohexane

☑ LONGEST CHAIN ☑ LOWEST LOCANTS ☑ ALPHABETICAL ORDER ☑ PUNCTUATION ☑ FORMULA COUNTED

CARD A09

Level: Challenge | Molecular Formula: $C_{10}H_{22}$

NUMBERED PARENT



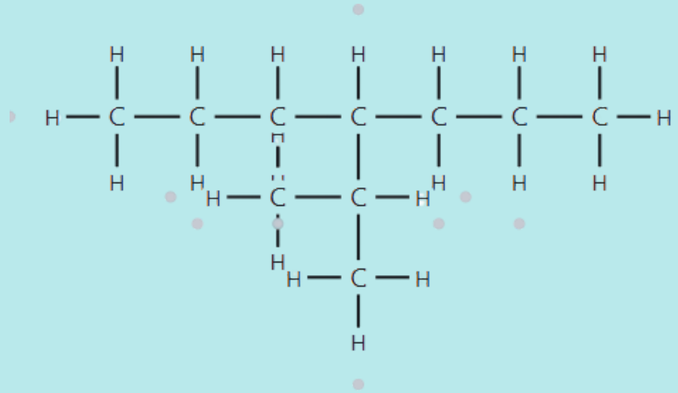
NAMING TRAIL

Parent: Longest chain = 6C → hexane
Numbering: Start from the top tip of the "Y" shape and trace down and to the right to give the branches the lowest possible numbers (2 and 3).
Substituents: Methyl on C2, Propyl on C3 → 2-methyl and 3-propyl

MY NAME

2-methyl-3-propylhexane

THE ALKANE CHECK



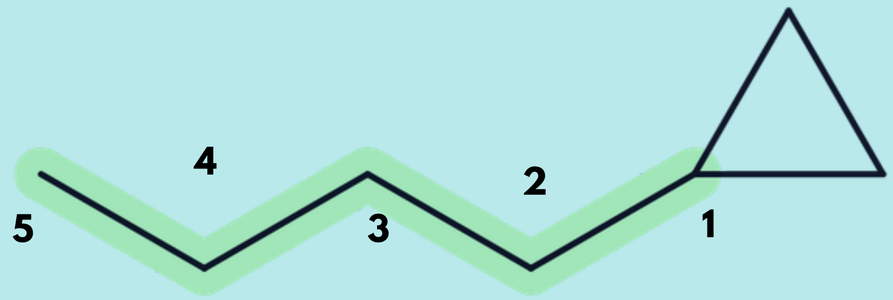
Returned Name: 2-methyl-3-propylhexane

☑ LONGEST CHAIN ☑ LOWEST LOCANTS ☑ ALPHABETICAL ORDER ☑ PUNCTUATION ☑ FORMULA COUNTED

CARD A10

Level: Challenge | Molecular Formula: C_7H_{14}

NUMBERED PARENT



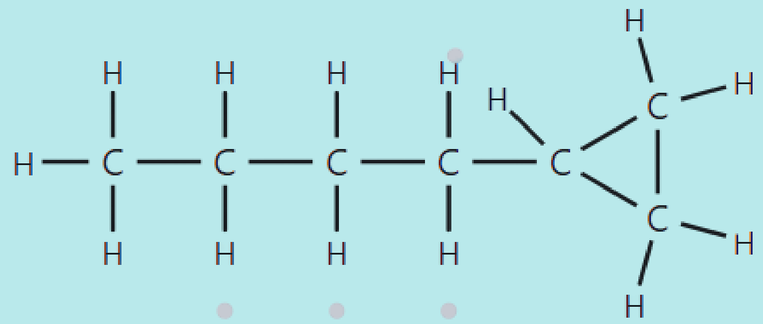
NAMING TRAIL

Parent: Longest chain = 5C → pentane
Numbering: left → right gives 5; right → left gives 1.
First point of difference: 1 < 5, so number from the right.
Substituents: Cyclopropyl ring on C1 → 1-cyclopropyl

MY NAME

1-cyclopropylpentane

THE ALKANE CHECK



Returned Name: butylcyclopropane

☑ LONGEST CHAIN ☑ LOWEST LOCANTS ☑ ALPHABETICAL ORDER ☑ PUNCTUATION ☑ FORMULA COUNTED

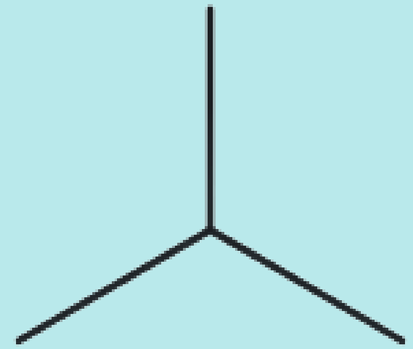
CARD B01

Level: Foundation | Given Name: 2-methylpropane

NAME DECODED

Parent: propane → 3C
Substituents: Methyl on C2
Expected Formula: 3 + 1 = 4C; alkane → C_4H_{10}

BOND LINE STRUCTURE



FORMULA CHECK

Counted from the drawing: C_4H_{10}
Alkane: C_nH_{2n+2} = C_4H_{10} ✓

BACK CHECK

Returned by The Alkane: 2-methylpropane
Same as given name ✓

☑ PARENT LENGTH MATCH ☑ SUBSTITUENT POSITION MATCH ☑ FORMULA AGREE ☑ BACK CHECK RECORDED

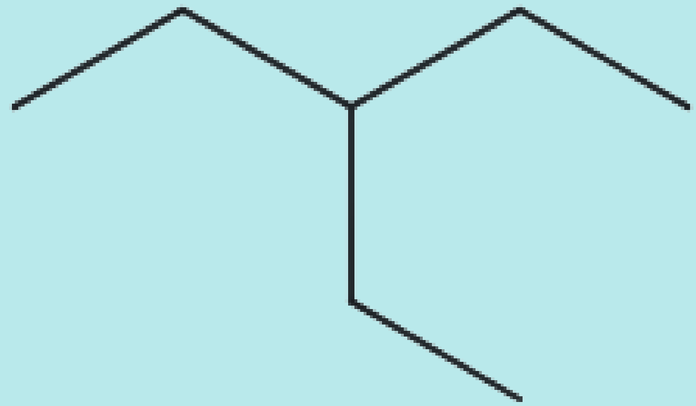
CARD B02

Level: Foundation | Given Name: 3-ethylpentane

NAME DECODED

Parent: pentane → 5C
Substituents: Ethyl on C3
Expected Formula: 5 + 2 = 7C; alkane → C_7H_{16}

BOND LINE STRUCTURE



FORMULA CHECK

Counted from the drawing: C_7H_{16}
Alkane: C_nH_{2n+2} = C_7H_{16} ✓

BACK CHECK

Returned by The Alkane: 3-ethylpentane
Same as given name ✓

☑ PARENT LENGTH MATCH ☑ SUBSTITUENT POSITION MATCH ☑ FORMULA AGREE ☑ BACK CHECK RECORDED

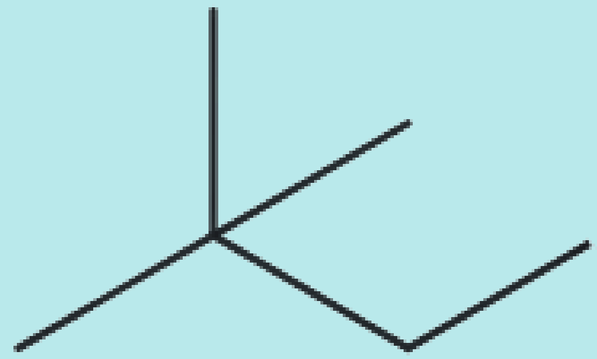
CARD B03

Level: Foundation | Given Name: 2,2-dimethylbutane

NAME DECODED

Parent: butane → 4C
Substituents: Two methyl groups on C2
Expected Formula: 4 + 2 = 6C; alkane → C_6H_{14}

BOND LINE STRUCTURE



FORMULA CHECK

Counted from the drawing: C_6H_{14}
Alkane: C_nH_{2n+2} = C_6H_{14} ✓

BACK CHECK

Returned by The Alkane: 2,2-dimethylbutane
Same as given name ✓

☑ PARENT LENGTH MATCH ☑ SUBSTITUENT POSITION MATCH ☑ FORMULA AGREE ☑ BACK CHECK RECORDED

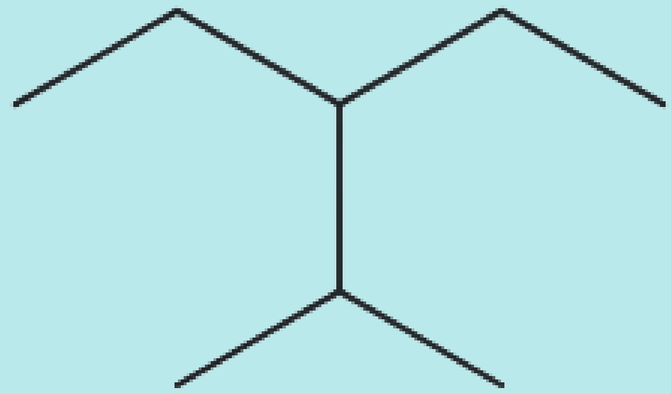
CARD B04

Level: Developing | Given Name: 3-ethyl-2-methylpentane

NAME DECODED

Parent: pentane → 5C
Substituents: Ethyl on C3, Methyl on C2
Expected Formula: 5 + 2 + 1 = 8C; alkane → C_8H_{18}

BOND LINE STRUCTURE



FORMULA CHECK

Counted from the drawing: C_8H_{18}
Alkane: C_nH_{2n+2} = C_8H_{18} ✓

BACK CHECK

Returned by The Alkane: 3-ethyl-2-methylpentane
Same as given name ✓

☑ PARENT LENGTH MATCH ☑ SUBSTITUENT POSITION MATCH ☑ FORMULA AGREE ☑ BACK CHECK RECORDED

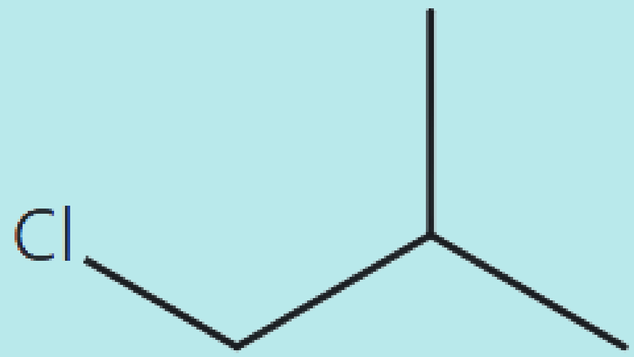
CARD B05

Level: Developing | Given Name: 1-chloro-2-methylpropane

NAME DECODED

Parent: propane → 3C
Substituents: Chloro on C1, Methyl on C2
Expected Formula: 3 + 1 = 4C; alkane → C_4H_{10}; Cl replaces one H → C_4H_9Cl

BOND LINE STRUCTURE



FORMULA CHECK

Counted from the drawing: C_4H_9Cl
Alkane: C_nH_{2n+2} = C_4H_{10}; minus one H for Cl = C_4H_9Cl ✓

BACK CHECK

Returned by The Alkane: 1-chloro-2-methylpropane
Same as given name ✓

☑ PARENT LENGTH MATCH ☑ SUBSTITUENT POSITION MATCH ☑ FORMULA AGREE ☑ BACK CHECK RECORDED

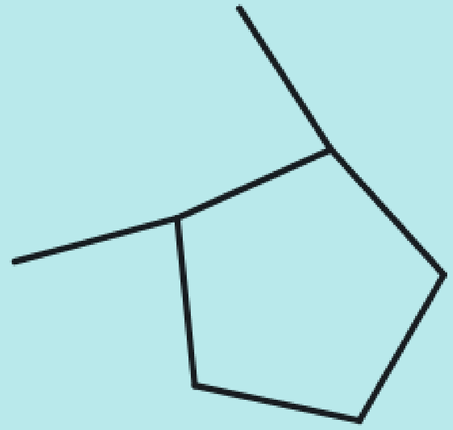
CARD B06

Level: Developing | Given Name: 1,2-dimethylcyclopentane

NAME DECODED

Parent: cyclopentane → 5C ring
Substituents: Methyl on C1, C2
Expected Formula: 5 + 2 = 7C; one ring → C_7H_{14}

BOND LINE STRUCTURE



FORMULA CHECK

Counted from the drawing: C_7H_{14}
One ring: C_nH_{2n} = C_7H_{14} ✓

BACK CHECK

Returned by The Alkane: 1,2-dimethylcyclopentane
Same as given name ✓

☑ PARENT LENGTH MATCH ☑ SUBSTITUENT POSITION MATCH ☑ FORMULA AGREE ☑ BACK CHECK RECORDED

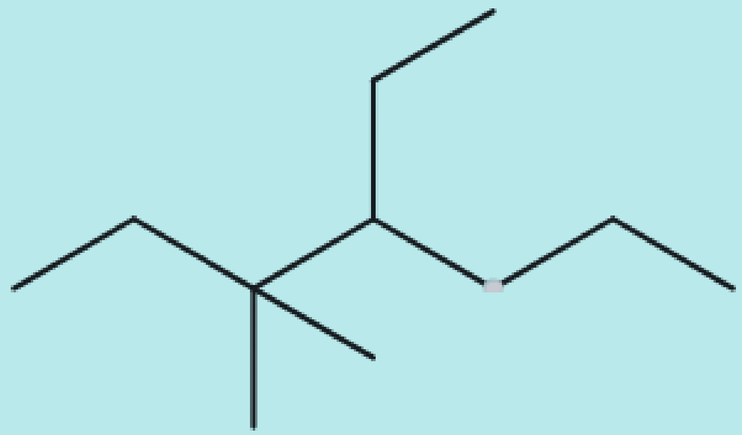
CARD B07

Level: Developing | Given Name: 4-ethyl-3,3-dimethylheptane

NAME DECODED

Parent: heptane → 7C
Substituents: Ethyl on C4, two methyl on C3
Expected Formula: 7 + 2 + 2 = 11C;
alkane → $C_{11}H_{24}$

BOND LINE STRUCTURE



FORMULA CHECK

Counted from the drawing: $C_{11}H_{24}$
Alkane: C_nH_{2n+2} = $C_{11}H_{24}$ ✓

BACK CHECK

Returned by The Alkane: 4-ethyl-3,3-dimethylheptane
Same as given name ✓

☑ PARENT LENGTH MATCH ☑ SUBSTITUENT POSITION MATCH ☑ FORMULA AGREE ☑ BACK CHECK RECORDED

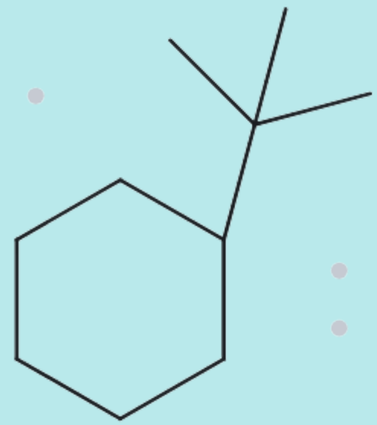
CARD B08

Level: Challenge | Given Name: tert-butylcyclohexane

NAME DECODED

Parent: cyclohexane → 6C ring
Substituents: tert butyl group on C1
Expected Formula: 6 + 4 = 10C; one ring → $C_{10}H_{20}$

BOND LINE STRUCTURE



FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$
One ring: C_nH_{2n} = $C_{10}H_{20}$ ✓

BACK CHECK

Returned by The Alkane: (1,1-dimethylethyl)cyclohexane
Same as given name ✓

☑ PARENT LENGTH MATCH ☑ SUBSTITUENT POSITION MATCH ☑ FORMULA AGREE ☑ BACK CHECK RECORDED

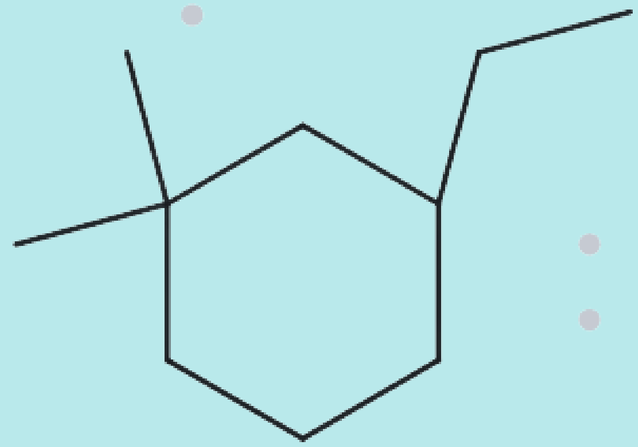
CARD B09

Level: Challenge | Given Name: 3-ethyl-1,1- dimethylcyclohexane

NAME DECODED

Parent: cyclohexane → 6C ring
Substituents: Ethyl on C3, two Methyl on C1
Expected Formula: 6 + 2 + 2 = 10C;
one ring → $C_{10}H_{20}$

BOND LINE STRUCTURE



FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$
One ring: C_nH_{2n} = $C_{10}H_{20}$ ✓

BACK CHECK

Returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane
Same as given name ✓

☑ PARENT LENGTH MATCH ☑ SUBSTITUENT POSITION MATCH ☑ FORMULA AGREE ☑ BACK CHECK RECORDED

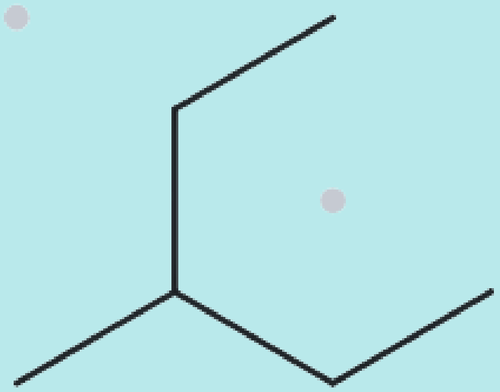
CARD B10

Level: Challenge | Given Name: 2-ethylbutane

NAME DECODED

Parent: butane → 4C
Substituents: Ethyl on C2
Expected Formula: 4 + 2 = 6C; alkane → C_6H_{14}

BOND LINE STRUCTURE



FORMULA CHECK

Counted from the drawing: C_6H_{14}
Alkane: C_nH_{2n+2} = C_6H_{14} ✓

BACK CHECK

Returned by The Alkane: 3-methylpentane
Same as given name ✓

☑ PARENT LENGTH MATCH ☑ SUBSTITUENT POSITION MATCH ☑ FORMULA AGREE ☑ BACK CHECK RECORDED